A proof of Hardin's empirical column formulas for OEIS A188866 via ranges of lazy walks, formalized in Lean 4

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Abstract

Let $T(n,k)$ be the number of $n\times k$ binary arrays without the pattern 0 1 vertically, diagonally or antidiagonally (OEIS A188866 [1]). R. H. Hardin recorded eight column formulas, from $T(n,1) = n+1$ to $T(n,8) = 2187n - 3495$ for $n > 5$, which are labelled "Empirical" in the OEIS. We prove the uniform closed form $T(n,k) = 3^{k-1}(n+1) - R_{k-1}$ for all $k \geq 1$ and $n \geq k-2$. Here R_L is the total range of all 3^L lazy walks of length L. Evaluating $R_0, \ldots, R_7$ yields all eight of Hardin's formulas, each on exactly Hardin's range of validity. The whole development is machine-checked in Lean 4 with Mathlib [6]: no `sorry`, and only the axioms `propext`, `Classical.choice` and `Quot.sound`.

1 Introduction

OEIS A188866 (R. H. Hardin, 2011) tabulates $T(n,k)$, the number of $n \times k$ binary arrays in which a 0 at position (i,j) forces 0 at $(i+1, j-1)$, $(i+1, j)$ and $(i+1, j+1)$ whenever those cells exist. The entry lists the formulas

$$\begin{array}{lll} T(n,1) = n+1, & T(n,2) = 3n+1, & T(n,3) = 9n-1, \\ T(n,4) = 27n-13 \ (n>1), & T(n,5) = 81n-65 \ (n>2), & T(n,6) = 243n-265 \ (n>3), \\ T(n,7) = 729n-987 \ (n>4), & T(n,8) = 2187n-3495 \ (n>5), & \end{array}$$

all marked "Empirical". We prove them, together with a single formula that covers every column.

2 Related work

The table A188866 and the formulas were contributed to the OEIS by R. H. Hardin in 2011 [1]. Columns $k = 4, \ldots, 8$ are the separate entries A188861–A188865 [2], which repeat the corresponding empirical formulas. A. Howroyd (2017) gave a bijection between these arrays and strings over $\{0, \ldots, n\}$ of length k whose adjacent letters differ by at most one [3]. These strings are the *smooth words* of Knopfmacher, Mansour, Munagi and Prodinger [4], who give their generating function (Theorem 2.2) and an explicit trigonometric formula for their number (Theorem 2.4). **In principle, therefore, the column formulas follow from Howroyd's bijection combined with [4].** Our contribution is a direct, elementary proof that gives a uniform closed form with constant term R_{k-1}, the total range of lazy walks, together with what is to our knowledge the first machine-checked verification. Together these turn the OEIS's "Empirical" column formulas for $k = 1, \ldots, 8$ into proved results. Neither R_L nor $3^L - R_L$ appeared in an OEIS search on October 12, 2026. The problem is not in the `formal-conjectures` repository [5], so our Lean definition is transcribed directly from the OEIS entry. The formalization uses Lean 4 [7] and Mathlib [6].

3 Statement

For $s \in \{-1,0,1\}^L$ put $S_0 = 0$ and $S_i = s_1 + \cdots + s_i$, and define

$$R_L = \sum_{s \in \{-1,0,1\}^L} \Big(\max_{0 \le i \le L} S_i - \min_{0 \le i \le L} S_i \Big).$$

Then $R_0, \ldots, R_9 = 0, 2, 10, 40, 146, 508, 1716, 5682, 18546, 59884$.

Theorem 1. *For all $k \ge 1$ and $n \ge k-2$, $T(n,k) = 3^{k-1}(n+1) - R_{k-1}$.*

Corollary 2. *All eight of Hardin's column formulas hold, each on the range stated in the OEIS.*

4 Proof overview

Step 1 (arrays and smooth words). Taking $j' = j$ shows that each column is a block of 1s followed by 0s. Let $t_j \in \{0, \ldots, n\}$ be the height of column j. The diagonal conditions are then equivalent to $|t_j - t_{j+1}| \le 1$. So $T(n,k) = W(n, k-1)$, where $W(M,L)$ counts words $t \in \{0, \ldots, M\}^{L+1}$ whose adjacent letters differ by at most one.

Step 2 (fibres over step sequences). A smooth word is determined by its step sequence $s \in \{-1,0,1\}^L$ and its first letter x, since $t_i = x + S_i$. The word stays in $[0, M]$ exactly when $x \in [-\min S,\ M - \max S]$. Because $\max S - \min S \le L \le M+1$, this interval contains exactly $M + 1 - (\max S - \min S)$ integers. In particular it is never too short for the count to be correct, a boundary case that must be handled explicitly. Summing over all 3^L step sequences gives $W(M,L) = 3^L(M+1) - R_L$.

Step 3. Kernel evaluation of $R_0, \ldots, R_7$ gives Corollary 2.

5 Formalization

Table 1 lists each mathematical result next to its Lean name. All declarations are in namespace `NovelProof`.

Lean name (file)	Statement
`Valid`, `T` (`Arrays.lean`)	the OEIS avoidance condition; $T(n,k)$
`WValid`, `W`, `R` (`Words.lean`)	smooth words; $W(M,L)$; R_L
`fiber_card`, `hi_add_lo_le`	Step 2 fibre count; range $\le L$
`W_formula`	$L \le M+1 \Rightarrow W(M,L) = 3^L(M+1) - R_L$
`T_eq_W`	$T(n, L+1) = W(n,L)$ (Step 1)
`T_formula`	Theorem 1
`R_values` (`Corollaries.lean`)	$R_0, \ldots, R_7$
`T_col1`, …, `T_col8`	Corollary 2

Table 1: Theorems and their Lean names.

6 Reproducibility

- Source code: `https://github.com/het4rk/OEIS-A188866-proof`. The Lean project is in `NovelProof/`.

- Toolchain: `leanprover/lean4:v4.35.0-rc4`. Mathlib: tag `v4.35.0-rc4`, commit `1f414401f69059aa7eead47b53ee40bd38455eeb`, pinned in `lake-manifest.json`.
- Build:

```
cd NovelProof
lake exe cache get
lake build
lake env lean Axioms.lean
```

- Axioms: `T_eq_W`, `T_formula`, `T_col4` and `T_col8` each depend only on `propext`, `Classical.choice` and `Quot.sound`.
- Attestation: the directory `verification/` contains the fresh-clone build log, the axiom output and `VERIFICATION.md`, which records the commit, the toolchain and SHA-256 hashes of the Lean sources. Continuous integration (`https://github.com/het4rk/OEIS-A188866-proof/actions/workflows/lean.yml`) rebuilds the project on every push. It rejects any `sorry` and any axiom outside the three above, and uploads the build log and axiom report as artifacts.

7 Limitations

We give no closed form or generating function for R_L. Hardin's empirical row recurrences (fixed n, varying k) are not formalized. They follow from the transfer-matrix and Chebyshev description [4]. The novelty is modest: the result settles a family of empirical OEIS formulas and identifies their constant term.

References

[1] R. H. Hardin, *A188866: $T(n,k)$ is the number of $n \times k$ binary arrays without the pattern* 0 1 *diagonally, vertically or antidiagonally*, The On-Line Encyclopedia of Integer Sequences, 2011, `https://oeis.org/A188866`.

[2] R. H. Hardin, *A188861, A188862, A188863, A188864, A188865* (columns $k = 4, \ldots, 8$ of A188866), The On-Line Encyclopedia of Integer Sequences, 2011, `https://oeis.org/A188861`.

[3] A. Howroyd, *Bijection with* 0..n *strings of length* k, 2017, linked from A188866, `https://oeis.org/A188866/a188866.txt`.

[4] A. Knopfmacher, T. Mansour, A. Munagi, H. Prodinger, *Smooth words and Chebyshev polynomials*, arXiv:0809.0551, 2008.

[5] Google DeepMind, *formal-conjectures*, `https://github.com/google-deepmind/formal-conjectures`.

[6] The mathlib Community, *The Lean mathematical library*, in Proc. CPP 2020, ACM, 367–381.

[7] L. de Moura, S. Ullrich, *The Lean 4 theorem prover and programming language*, in CADE-28, LNCS 12699, Springer, 2021, 625–635.